SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Concept Mapping

COURSE NAME: Data Structures **COURSE CODE: CSA0308**

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COURSE OUTCOME COVERED

CO1: *Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* (BL2, BL3)

Assessment Tool Weightage: 35%

Questions:2 **EACH QUESTIONS CARRIES:** 10 MARKS **Total Marks:**20 **Duration:** ONE HOUR

Q1 A smart healthcare system processes large volumes of patient data such as temperature readings, medical history, and diagnostic details. Construct a concept map linking the progression from Data → Data Object → Data Structure → Abstract Data Type (ADT) → Implementation in C. Identify relevant sub-concepts such as structures, pointers, and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how each level contributes to efficient data organization, processing, and improved system performance.

Q2 A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non-Primitive Data Structures → Linear → Non-Linear Structures. Include examples such as arrays, linked lists, trees, and graphs, and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

Code of Conduct:

I Dhavanesh S S (1972260011) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

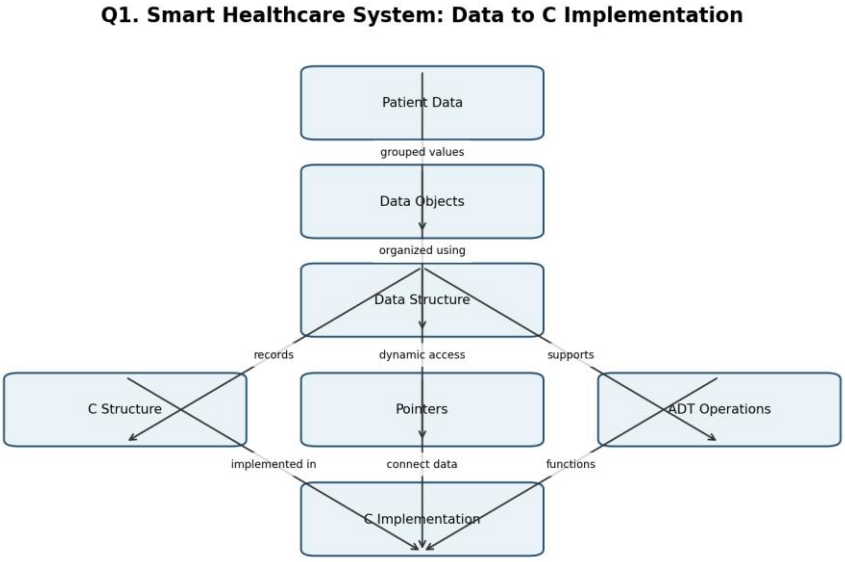
RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

CO1 – AT1: ANSWERS

Q1. Smart Healthcare System – Data → Data Object → Data Structure → ADT → C Implementation

A smart healthcare system handles temperature readings, medical history and diagnostic details. These values are grouped into data objects and organized with suitable data structures. An Abstract Data Type (ADT) defines the operations that can be performed, while C structures, pointers and functions provide an implementation.



Explanation and justification

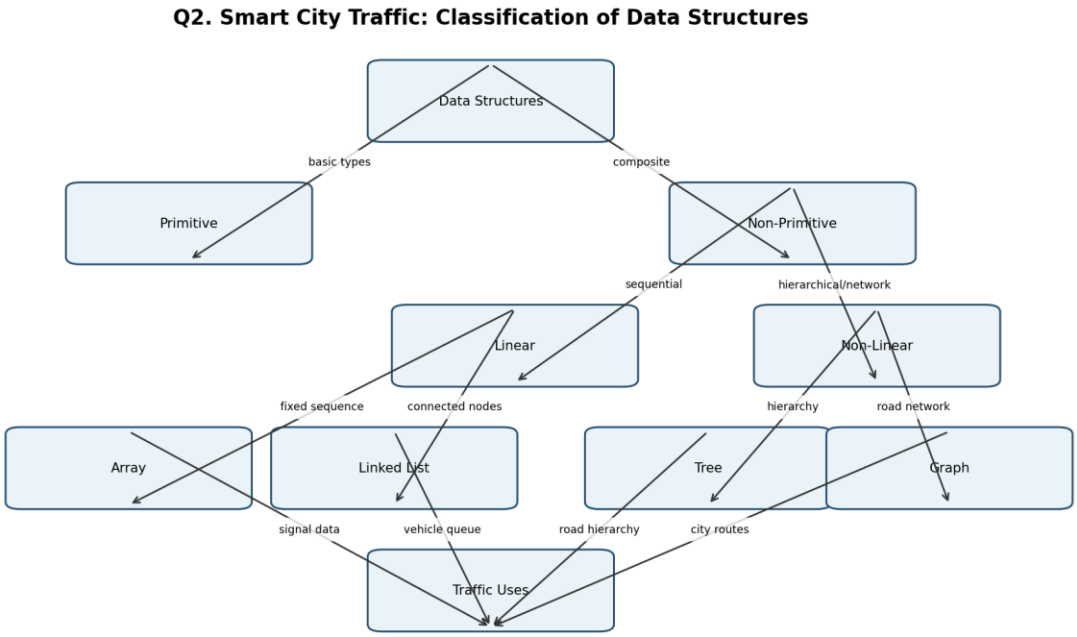
- Data: Raw values such as temperature, pulse rate, patient ID and diagnostic observations.
- Data object: Related values belonging to one patient are grouped as a single logical record.
- Data structure: A structure can combine fields of different data types; an array or linked list can store multiple patient records.
- ADT: Specifies permitted operations such as insert patient, update readings, search patient and display history, independent of implementation.
- C implementation: struct defines a record, pointers refer to records and dynamically allocated nodes, and functions implement the ADT operations.
- Efficiency: Choosing the right organization improves retrieval and updates. Direct array indexing is O(1); searching an unsorted collection is O(n); linked-list insertion at a known position can be O(1).

Conclusion

The progression from raw data to data objects, data structures, ADTs and C implementation separates the information from the operations performed on it. This supports organized patient records, maintainability and efficient processing.

Q2. Smart City Traffic Management – Data Structure Classification

A smart city traffic system manages vehicle movement, road connections and signal timing. Data structures are classified into primitive and non-primitive types. Non-primitive structures are further classified as linear and non-linear based on how elements are related.



Explanation and traffic applications

- Primitive data structures: int, char, float and bool store basic values such as vehicle count, signal state and speed.
- Non-primitive data structures: Organize multiple values and relationships; examples include arrays, linked lists, trees and graphs.
- Linear structures: Arrays store signal timings or sensor readings by index. Linked lists can represent changing vehicle queues.
- Non-linear structures: Trees can represent hierarchical road or control-zone information. Graphs represent intersections as vertices and roads as edges; weights can represent distance or travel time.
- Efficiency: Array access by index is O(1); traversing a list is O(n). Graphs model interconnected roads naturally and support shortest-path algorithms such as Dijkstra's algorithm when edge weights are non-negative.

Conclusion

Primitive structures store individual values, while non-primitive structures organize collections. Linear structures suit sequential data; trees and graphs suit hierarchical and network relationships. A weighted graph is especially suitable for city-road connectivity.